Thammana Kumar Raja

Machine Learning Engineer

+91-7036110994 | kumarrajathammana1918@gmail.com | Bengaluru / Hyderabad (Open to Relocate)

LinkedIn: linkedin.com/in/kumar-raja-3ba124235 | GitHub: github.com/kumarrajathammana918

CAREER OBJECTIVE

Machine Learning Engineer with 1 year of experience building data-driven and GenAI solutions. Skilled in RAG pipelines, agentic AI workflows, classical ML, and MLOps on Azure. Seeking to leverage expertise in LangChain, LangGraph, Azure OpenAI, and responsible AI practices to deliver measurable business impact.

EDUCATION

GITAM Institute of Technology — Jan 2021 – April 2025
Bachelor of Technology in Computer Science — *Visakhapatnam*

Sri Gayatri Junior College — June 2019 – May 2021
Intermediate - MPC — *Visakhapatnam*

Vignan Vihara School — Jan 2018 – May 2019
SSC — *Visakhapatnam*

TECHNICAL SKILLS

Programming Languages: Python, SQL, Git, HTML/CSS

Machine Learning: scikit-learn, XGBoost, LightGBM, SHAP, MLflow, model evaluation (AUC, F1), imbalance handling

NLP and Deep Learning: Prompt engineering, structured outputs (JSON), context windows, VGG16, CNN, OpenCV

Generative AI: RAG, Embeddings, Hybrid retrieval, Chunking strategies, Citations, Guardrails

Agentic AI: LangGraph, LangChain, LlamaIndex, Tool/function calling, ReAct-style orchestration, Pydantic

Data and Libraries: Pandas, NumPy, Matplotlib, FastAPI, EDA, Feature engineering

Cloud and MLOps: Azure OpenAI, Azure AI Search, MLflow, experiment tracking, retraining pipelines

WORK EXPERIENCE

Machine Learning Engineer

VisionQuest Solutions Pvt. Ltd. | Bengaluru, India   Aug 2025 – Present

- Designed and deployed GenAI solutions using RAG pipelines with grounded responses, citations, and fallback behavior, improving answer faithfulness by ~35%.
- Built agentic AI workflows using LangGraph with tool/function calling, multi-step planning, and verification for complex business tasks.
- Engineered structured prompt templates for consistent JSON outputs and safe responses with content guardrails, reducing hallucination risk.
- Developed classical ML classification models (XGBoost, LightGBM) with stratified cross-validation, leakage checks, and threshold tuning.
- Delivered SHAP-based model explainability and error analysis, translating outputs into measurable retention and campaign strategies.
- Implemented MLflow experiment tracking and reusable training pipelines, reducing re-training setup time.
- Created data quality monitoring signals for drift, missingness, and performance regression with defined retraining triggers.
- Collaborated cross-functionally with stakeholders to define success metrics, acceptance criteria, and evaluation sets.
- Produced audit-ready documentation for datasets, prompts, evaluations, and model versioning decisions.

PROJECTS

Content-Based Image Retrieval for Brain Tumor Detection | Python, Deep Learning, VGG16, CNN, Gradio

- Developed a CBIR system for efficient and accurate retrieval of similar brain MRI images for tumor analysis.

- Employed a pre-trained VGG16 CNN to extract high-dimensional feature vectors from MRI scans for comparison.
- Stored feature vectors in a database and retrieved similar images using Euclidean distance similarity metrics.
- Implemented image preprocessing pipeline and integrated an interactive Gradio interface for ranked result display.

GenAI Patient Support Knowledge Assistant (RAG) **| Python, Azure OpenAI, LangChain, Azure AI Search, Guardrails**

- Designed a RAG pipeline with role-based retrieval and metadata filters, preventing cross-department data leakage.
- Created safe prompt templates with medical disclaimers and escalation routing for sensitive queries.
- Built an evaluation set of 150+ support intents to track retrieval accuracy and refusal correctness across iterations.
- Delivered structured JSON outputs powering FAQ links, citations, and next-step actions in the support UI.

Customer Churn Classification and Retention Targeting **| Python, scikit-learn, XGBoost, LightGBM, SHAP, MLflow, SQL**

- Built a churn prediction model achieving strong AUC using XGBoost with stratified cross-validation and leakage checks.
- Handled class imbalance and calibrated decision thresholds to optimize business precision and recall trade-offs.
- Delivered SHAP-based feature explainability and error slice analysis for segment-level campaign targeting.
- Packaged a batch scoring pipeline and tracked all experiments with MLflow for reproducibility and governance.